

ONS TAHA

AI_Data Engineer

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PROFILE

- Results-driven Software, Data & AI Engineer with a strong record of architecting scalable backend systems, enterprise data platforms, and intelligent AI solutions that deliver measurable business transformation. Specialized in C#/.NET (Clean Architecture & Domain-Driven Design), large-scale data warehousing, ELT/ETL orchestration, and performance-optimized system design within high-demand environments.
- Architect of end-to-end AI systems—including OCR pipelines, predictive analytics models, NLP clustering frameworks, Retrieval-Augmented Generation (RAG) platforms, and LLM-powered enterprise assistants—integrating structured and unstructured data into production-ready solutions that automate workflows, enhance decision intelligence, and improve operational efficiency.
- Bridges business strategy with advanced engineering execution by translating complex domain challenges into scalable, reliable architectures. Recognized for driving data-informed decision-making, optimizing system performance, and delivering technology solutions that create sustainable organizational impact.

WORK EXPERIENCE

AAU Company Dec 2025 – Present

Software Engineer

- Designed and developed scalable backend systems using C# and ASP.NET Core, following Clean Architecture and Domain-Driven Design (DDD) principles.
- Designed modular applications using ABP layered architecture (Domain, Application, Entity Framework Core, HttpApi).
- Modeled complex business domains using Entities, Value Objects, Aggregates, and Domain Services to ensure maintainable and scalable solutions.
- Implemented Application Services and DTO mapping using AutoMapper following ABP best practices.
- Managed database operations using Entity Framework Core with PostgreSQL, including migrations, indexing, and query optimization.
- Developed RESTful APIs with proper validation, exception handling, and standardized response wrapping.
- Architected high-performance backend systems leveraging Redis for distributed caching and task orchestration and MinIO for scalable object storage, enabling efficient handling of large-scale document
- Designed middleware to manage request lifecycle stages including inspection, preprocessing, and controlled pipeline execution.
- Built end-to-end authentication flows, including registration, login, and OTP verification, integrating secure token handling and validation logic.

QI Card International Smart Card (ISC) Feb 2025 – Dec 2025

AI/Data Engineer

- Designed, deployed, and maintained machine learning and deep learning models for real-world business use cases with a focus on reliability and scalability
- OCR System: Collected and labeled real-world document images, fine-tuned YOLO for page layout segmentation (header/body/footer), designed image preprocessing pipelines, and fine-tuned PaddleOCR models for high-accuracy multilingual text extraction in production OCR systems.
 - Built backend workflows for handling, storing, and retrieving document assets using scalable storage systems.
- AI Chatbot / Agent: Built an AI agent to analyze OCR output and document metadata, enabling automated classification, intelligent decision-making, and assisted response generation within document processing workflows.
- RAG & Knowledge Systems: Implemented a Retrieval-Augmented Generation (RAG) system that indexes enterprise documents, retrieves contextually relevant records per query, and serves as a smart organizational knowledge warehouse.
- Mayan EDMS (AI Workflow Extension): modified and extended Mayan EDMS source code to integrate custom AI modules, expose new AI workflow actions in the UI, and deliver full end-to-end integration across frontend, backend logic, external AI services, and database persistence—enabling document classification, metadata auto-filling, intelligent tagging, and AI-assisted reply drafting within native Mayan workflows.
- Architected and deployed containerized AI microservices using Docker Compose (PostgreSQL, Redis, RabbitMQ, FastAPI), supporting secure and offline-ready environments for regulated financial operations.
- Architected and implemented an LLM-based chatbot using Retrieval-Augmented Generation (RAG) for a banking client, integrating structured and unstructured knowledge sources, enforcing domain constraints, and ensuring the response quality, usage, and operational impact all in context.
- Speech-to-Text Optimization: Converted Whisper models from PyTorch to ONNX to improve inference performance and enable seamless integration with C#/.NET-based systems.

EarthLink Company APR 2023 – Mar 2025

Data Engineer

- owned end-to-end design and operation of ELT/ETL pipelines, integrating data from diverse source systems into a centralized data warehouse while enforcing data quality, consistency, and reliability standards.
- Designed and implemented a complete data warehouse architecture in ClickHouse, including fact and dimension modeling (star/snowflake schemas), optimized partitioning, ordering, indexing, and aggregation strategies to support large-scale analytical workloads, high-concurrency queries, and low-latency reporting.
- Orchestrated and scheduled data pipelines using workflow orchestration tools (Airflow), managing dependencies, retries, backfills, and SLA-aligned execution to ensure reliable and timely data availability.
- Worked side by side with stakeholders across multiple business domains—including Customer Experience, Media & Movies, Marketing, Support, HR, and Network Operations—to understand business requirements and translate them into scalable, data-driven solutions.
- Acted as a key bridge between business and technical teams by owning the end-to-end translation of business objectives into robust analytical and reporting architectures, driving data-driven decision-making at multiple organizational levels.
- Extracted, analyzed, and modeled data from disparate systems to identify key trends, patterns, and performance metrics, supporting strategic and operational decision-making.
- Designed and delivered advanced dashboards and reports using Power BI and Grafana, translating complex data into clear, actionable insights for executives and operational teams.
- Developed Python automation scripts with business rules and integrated the data with external platforms via APIs, using warehouse-derived insights to orchestrate automated operational actions and ensuring seamless operations.
- Designed and implemented applied AI solutions across operational and media domains, including:
 - Predictive Analytics (Supervised ML) Implemented, trained, and evaluated predictive machine learning models using monitoring data (Zabbix), temporal signals (days, events, holidays), and traffic features to forecast future workload and anticipate potential device or user failures, validating performance with appropriate accuracy and evaluation metrics.
 - Designed, trained, and evaluated an NLP-based unsupervised learning pipeline, applying text vectorization on movie descriptions and clustering techniques to identify semantic similarity and content groupings, with model quality assessed using appropriate evaluation metrics.

We Datum Software Solution Company 5 JAN – 15 APR 2023

Digital Marketing

(Marketing planning department)

Search Engine Optimization (SEO) and search engine marketing Event Planning

(Campaign Planning and Execution Media Analysis Brand Awareness

PROJECTS

- Trained model for movies work as recommendation system and chat box at the same time (EarthLink Company)
- Estimated the zabbix monitoring system issues use deep learning techniques (EarthLink Company)
- Diagnosis the level of hypertension using machine learning technique (university of Technology)

SKILLS

- **Soft Skills**

- Analytical Thinking
- Problem Solving
- Cross-functional Collaboration
- Stakeholder Communication
- Requirements Analysis
- Business-Technical Bridging

- **Programming language :**

- Python
- C# (.net)
- JavaScript

- **Data Engineering & Architecture**

- End-to-End ELT / ETL Pipeline Design
- Data Warehouse Architecture & Modeling (Star / Snowflake Schemas)
- ClickHouse (Partitioning, Ordering, Indexing, Aggregations)
- Data Quality, Consistency & Reliability Enforcement
- Source-of-Truth Data Design

- **DataBases**

- SQL/mysql/clickhouse/postgres/mongo db
- Time series db promsql query use prometheus
- vector db : FAISS ,Milvus,Qdrant

- **Data analysis**

- Business-Oriented Analytical Decision Support
- Data-Driven Decision Enablement
- Advanced Data Analysis & Insight Generation
- Forecasting & Trend Analysis
- Statistical Modeling & Hypothesis Testing
- Quantitative & Statistical Analysis (Entropy-Based Distribution Analysis, KPI Modeling, Trend & Variance Analysis)

- **Data Visualization & Reporting**

- Power BI
- Grafana
- Business Intelligence Reporting
- KPI Design & Executive Dashboards

- **Artificial Intelligence:**

- Machine Learning (Supervised & Unsupervised)
- Deep learning
- NLP & Text Vectorization
- LLMs (Large Language Models)
- Feature engineering
- Model Training, Validation, Evaluation, and Optimization
- AI-driven application development (chatbots, smart search)
- clustering unsupervised machine learning

- **Computer Vision & OCR**

- Document layout detection (header / body / footer segmentation)
- Image preprocessing pipelines (denoising, binarization, normalization)
- OCR model training and optimization
- Dataset collection, labeling, and augmentation

- **BackEnd**

- Scalable backend system design using C#, ASP.NET Core, and ABP Framework
- Clean Architecture & Domain-Driven Design (DDD) implementation
- Modular layered architecture (Domain, Application, EF Core, HttpApi)
- RESTful API design with validation, middleware, and standardized response handling
- Advanced Entity Framework Core usage (migrations, indexing, query optimization)
- PostgreSQL integration and performance tuning
- Business domain modeling (Entities, Value Objects, Aggregates, Domain Services)
- DTO mapping & object transformation using AutoMapper
- Exception handling strategy & API reliability engineering

LANGUAGE

- Arabic
- English
- French

EDUCATION

- University Of Technology Bachelor of Biomedical Engineer (2017-2022)

CERTIFICATIONS & COURSES :

- kaizen process improvement certificate (by Earthlink 2025)
- Certificate AI cs50 provided from Harvard university 2024
- Certificate of data science by IEEE 2023
- Certificate of python programming by IEEE 2023
- Certificate supervised machine learning by IEEE2023
- Certificate of completion: SQL Fundamentals By TechDa 2023
- Certificate of completion: Python Beginner level By TechDa 2023
- Marketing course at Business by design and wedatum co. 2023
- Python course at Aldojo and code for Iraq
- Power course at aswar academy 2023
- CCTV course at aswar academy 2022
- Participate in bill state American university 2018